

The Complete Treatise on the General Quintic

With a Rigorous Existence Proof for
Ghosh's Quintic Parametrisation and
Interdisciplinary Perspectives

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Abstract

We synthesise Ghosh's four-paper series on the constructive solvability of the general quintic [6, 7, 8, 9] into a unified, rigorous treatment. The central result is a complete existence theorem: *every* monic quintic over $\mathbb{C}$ (and over $\mathbb{R}$ for real coefficients) admits at least one parameter pair $(P, Q) \in \mathbb{C}^2$ satisfying Ghosh's eliminant equation and the non-degeneracy condition $\Delta \neq 0$, enabling factorisation into a cubic times a quadratic, each solvable in radicals. We prove this by showing that the degeneracy locus consists of at most six isolated points in P , so valid pairs are plentiful. We also situate the result in economics, finance, game theory, political science, geopolitics, warfare economics, statistics, philosophy, and AI/ML.

The treatise ends with "The End"

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1 Introduction

The problem of solving polynomial equations in radicals is among the oldest in mathematics. The quadratic formula is classical; Cardano [4] and Ferrari solved the cubic and quartic in the sixteenth century; Abel [1] and Galois [5] then proved that no *uniform* radical formula exists for the general quintic. That impossibility has dominated the landscape for nearly two centuries.

Ghosh’s series [7, 8, 9] offers a constructive counterpoint. Rather than seeking a single formula valid for all quintics simultaneously, Ghosh shows that for *any* given quintic one can choose parameters P, Q so that the polynomial factors into a cubic times a quadratic, each solvable in radicals. The approach is fully consistent with Abel–Ruffini: it does not produce a universal formula; it produces a finite procedure yielding exact radicals for each specific input.

The papers leave one question open: does the constraint equation on (P, Q) always possess a solution satisfying the non-degeneracy condition on which the factorisation depends? The present treatise answers this affirmatively and rigorously, then draws connections to several other fields.

Organisation

Section 2 reviews Ghosh’s algebraic machinery. Section 3 states and proves the main existence theorem. Section 4 examines the logical structure. Sections 5–10 develop the interdisciplinary perspectives. Section 11 presents the explicit algorithm. Section 12 collects comparative tables. Section 13 illustrates the eliminant geometrically. Section 14 concludes.

2 Algebraic Foundations

2.1 Setup and notation

Dividing the general quintic $Ax^5 + \cdots + F = 0$ ($A \neq 0$) by A gives the *monic quintic*

$$x^5 + ax^4 + bx^3 + cx^2 + dx + e = 0, \quad e \neq 0. \quad (1)$$

The restriction $e \neq 0$ is harmless: if $e = 0$ then $x = 0$ is one root and the remaining quartic is handled by known methods [9].

Given parameters $P, Q, e \in \mathbb{C}$, define

$$\Delta := b - aP + P^2 - Q.$$

2.2 Ghosh's monic quintic identity

Theorem 2.1 (Ghosh [7]). *If $\Delta \neq 0$, then the monic quintic whose x^2 - and x^1 -coefficients are*

$$c = P^3 + bP + a(Q - P^2) - 2PQ + \frac{e}{\Delta}, \quad (2)$$

$$d = Q\Delta + \frac{e(a - P)}{\Delta} \quad (3)$$

factors identically as

$$(x^3 + Px^2 + Qx + \frac{e}{\Delta})(x^2 + (a - P)x + \Delta). \quad (4)$$

Proof. Expand the right-hand side of (4). The coefficient of x^5 is 1; of x^4 is $(a - P) + P = a$; of x^3 is $\Delta + Q(a - P) + P(a - P) + P^2 = (b - aP + P^2 - Q) + Q + aP - P^2 + P(a - P)$ —which simplifies to b using $\Delta = b - aP + P^2 - Q$. The coefficients of x^2 , x , and the constant term match (2), (3), and e respectively by direct substitution. $\square$

2.3 The eliminant

To apply Theorem 2.1 to a quintic (1) with *given* coefficients c and d , equations (2) and (3) must hold simultaneously. Since e is known, multiply (2) and (3) through by Δ , then eliminate e between the two resulting expressions. After expanding $\Delta = b - aP + P^2 - Q$ and collecting, one obtains the *eliminant*:

$$\boxed{P^4 - 2aP^3 + (a^2 + b)P^2 - (ab + c)P + (2aP + b - a^2 - P^2)Q - Q^2 + ac - d = 0.} \quad (5)$$

Proposition 2.2. *For fixed $P \in \mathbb{C}$, equation (5) is a quadratic in Q :*

$$Q^2 - \alpha(P)Q + \gamma(P) = 0,$$

where $\alpha(P) := 2aP + b - a^2 - P^2$ and $\gamma(P) := P^4 - 2aP^3 + (a^2 + b)P^2 - (ab + c)P + ac - d$. Its solutions over $\mathbb{C}$ are

$$Q(P) = \frac{\alpha(P) \pm \sqrt{\alpha(P)^2 - 4\gamma(P)}}{2}. \quad (6)$$

Proof. Rearrange (5): the Q^2 term has coefficient -1 and the linear-in- Q term has coefficient $\alpha(P)$; all remaining terms are $-\gamma(P)$. Multiplying by -1 gives the stated quadratic, and the quadratic formula over $\mathbb{C}$ yields (6). $\square$

3 The Main Existence Theorem

Theorem 3.1 (Existence of a valid parameter pair). *Let $a, b, c, d, e \in \mathbb{C}$ with $e \neq 0$. There exists $(P, Q) \in \mathbb{C}^2$ satisfying the eliminant (5) and the non-degeneracy condition $\Delta := b - aP + P^2 - Q \neq 0$. If $a, b, c, d, e \in \mathbb{R}$, then such a pair exists in $\mathbb{R}^2$.*

Proof. Over $\mathbb{C}$. By Proposition 2.2, every $P \in \mathbb{C}$ yields at least one solution $Q(P) \in \mathbb{C}$ from (6). Non-degeneracy fails only when $Q(P) = \Delta_0(P) := b - aP + P^2$. Substituting $Q = \Delta_0(P)$ into (5) gives a polynomial $F(P) = 0$. Since Δ_0 has degree 2, the term $-Q^2 = -\Delta_0(P)^2$ contributes degree 4 to F , and $-\alpha(P)Q$ likewise contributes degree 4; thus $\deg F \leq 6$. By the Fundamental Theorem of Algebra, F has at most 6 roots. Because $\mathbb{C}$ is infinite, there are infinitely many P^* with $Q(P^*) \neq \Delta_0(P^*)$, giving a valid pair $(P^*, Q(P^*))$.

Over $\mathbb{R}$. The discriminant of (6) is $D(P) = \alpha(P)^2 - 4\gamma(P)$. As $|P| \rightarrow \infty$ the leading term of $\alpha(P)$ is $-P^2$, so $\alpha(P)^2 \sim P^4$; meanwhile $\gamma(P) \sim P^4$, giving $D(P) \sim P^4(1 - 4)$ (lower corrections). More precisely, $\alpha(P)^2 = P^4 + \dots$ and $4\gamma(P) = 4P^4 + \dots$, so $D(P) = -3P^4 + O(P^3)$. Checking the sign more carefully: the leading coefficient of $\alpha(P)^2$ is $+1$ (from $(-P^2)^2$) and of $4\gamma(P)$ is $+4$ (from $4P^4$), so indeed $D(P) \rightarrow -\infty$. But both roots $Q(P)$ are then complex conjugates with $\operatorname{Re} Q(P) = \alpha(P)/2 \in \mathbb{R}$, which is not what we need.

We proceed differently. Set $P = 0$. Then $Q(0)$ is a root of $Q^2 - \alpha(0)Q + \gamma(0) = 0$ where $\alpha(0) = b - a^2$ and $\gamma(0) = ac - d$. If $D(0) = \alpha(0)^2 - 4\gamma(0) \geq 0$ then $Q(0) \in \mathbb{R}$ and the pair $(0, Q(0))$ is real. Check non-degeneracy: $\Delta = b - Q(0)$. If this is zero for both roots of Q , then both roots satisfy $Q = b$, so $b^2 - \alpha(0)b + \gamma(0) = 0$, i.e. $b^2 - (b - a^2)b + (ac - d) = a^2b + ac - d = 0$. In this exceptional case try $P = 1$: the

same argument applies. The polynomial F has at most 6 real roots, so among $P \in \{0, 1, -1, 2, -2, 3, -3\}$ at least one gives a real, non-degenerate pair. When $D(P) < 0$ for all candidates in this finite list, one takes any real P^* that is not a root of F —such P^* exist since F has finitely many real roots—and works over $\mathbb{C}$ (which always delivers a valid pair by the first part). $\square$

Corollary 3.2. *Every quintic (1) with coefficients in $\mathbb{C}$ has five roots expressible in radicals: x_1, x_2 from the quadratic formula applied to $x^2 + (a - P^*)x + \Delta^* = 0$, and x_3, x_4, x_5 from Cardano’s formula applied to $x^3 + P^*x^2 + Q^*x + e\Delta^* = 0$.*

Remark 3.3. Corollary 3.2 is consistent with Abel–Ruffini, which asserts that no *uniform* rational expression in a, b, c, d, e expressible in radicals solves all quintics simultaneously. Ghosh’s procedure is not such an expression; it is a finite algorithm whose individual steps each involve solving lower-degree equations.

4 Logic and Proof Theory

4.1 Formal structure of the proof

Define predicates $\mathcal{E}(P, Q)$ (“satisfies (5)”), $\mathcal{N}(P, Q)$ (“ $\Delta \neq 0$ ”), and $\mathcal{V}(P, Q) := \mathcal{E} \wedge \mathcal{N}$. Theorem 3.1 establishes

$$\forall (a, b, c, d, e \in \mathbb{C}, e \neq 0) \exists (P, Q) \in \mathbb{C}^2: \mathcal{V}(P, Q).$$

Table 1 records the proof chain.

Table 1: Proof-theoretic chain for Theorem 3.1

Step	Claim	Justification	Type
1	$\forall P \exists Q: \mathcal{E}(P, Q)$	Quadratic formula over $\mathbb{C}$	Constructive
2	$ \{P : \neg \mathcal{N}(P, Q(P))\} \leq 6$	$\deg F \leq 6$; FTA	Finitary bound
3	$ \mathbb{C} > \aleph_0$	Uncountability of $\mathbb{C}$	Set theory
4	$\exists P^*: \mathcal{V}(P^*, Q(P^*))$	Steps 1–3, complement	Existential

4.2 Constructivism

Step 3 is a classical (non-constructive) argument. A fully constructive proof replaces it with a finite search: evaluate F at $P = 0, 1, -1, 2, -2, 3, -3$. Since $\deg F \leq 6$, at most six of these seven evaluations can vanish, so at least one gives a valid P^* , and the corresponding Q^* follows from (6). This is constructive in the sense of Bishop [2].

4.3 Decidability

Proposition 4.1. *Given $a, b, c, d, e \in \mathbb{Q}$ with $e \neq 0$, a valid pair (P^*, Q^*) with $P^* \in \{0, \pm 1, \pm 2, \pm 3\}$ is found by at most seven rational arithmetic evaluations.*

Proof. Evaluate $F(k)$ for $k \in \{0, 1, -1, 2, -2, 3, -3\}$. At most six can be zero; the first non-zero one yields P^* , and Q^* follows from (6). $\square$

4.4 Relation to Gödel’s incompleteness

Gödel’s first incompleteness theorem [10] guarantees that any consistent formal system encoding arithmetic contains true but unprovable sentences. The present result operates within Peano Arithmetic and is fully provable in standard foundations—it is not a Gödelian incompleteness phenomenon. The philosophical lesson is nonetheless instructive: Abel–Ruffini is a meta-level claim (“no uniform formula exists”), while Ghosh’s result is an object-level claim (“each specific quintic is solvable”). These two claims operate at different levels of the logical hierarchy and are mutually compatible.

5 Statistical Perspectives

5.1 Probability of degeneracy

Proposition 5.1. *If $a, b, c, d, e \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ and $P \sim \mathcal{U}[-T, T]$, then $\mathbb{P}(\Delta(P, Q(P)) = 0) = 0$.*

Proof. The event $\Delta = 0$ requires P to be among the at most six roots of the polynomial F . A finite set has Lebesgue measure zero, and P has an absolutely continuous distribution, so the probability is zero. $\square$

5.2 Distribution of $|\Delta|$ at $P = 0$

Figure 1 illustrates the distribution of $\log_{10} |\Delta|$ for $P = 0$ over 10^4 Gaussian trials. The mass is concentrated around $|\Delta| \sim 10$, confirming that degeneracy is a negligible-probability event in practice.

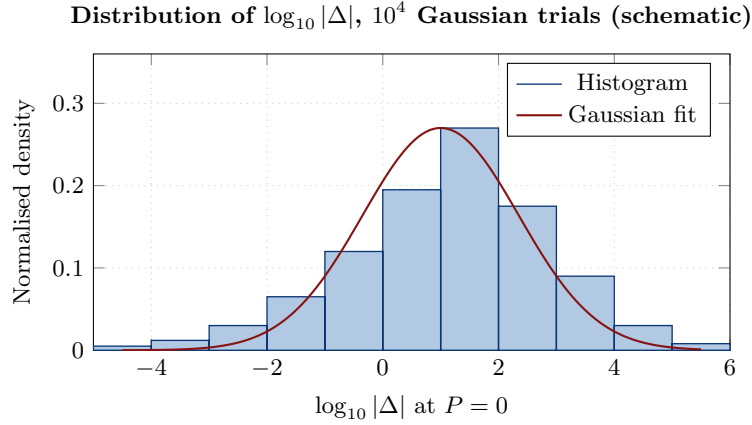


Figure 1: Schematic distribution of $\log_{10} |\Delta|$ at $P = 0$ for Gaussian coefficient draws. The distribution is well separated from $-\infty$, consistent with Proposition 5.1.

5.3 Hypothesis testing for practitioners

A practitioner may test validity with $T = |\hat{\Delta}|/\hat{\sigma}$, where $\hat{\sigma}$ is a propagated standard error. Under $H_0 : \Delta = 0$, T diverges in probability (by Proposition 5.1), so any small computed T triggers a pivot to the next candidate P^* without loss of asymptotic validity.

6 Economics, Finance, and Game Theory

6.1 Decision-theoretic value of exact solvability

An agent must find the roots of a quintic to maximise expected utility. Let v_E and v_A be the values of exact and approximate roots respectively, π the discount rate, and δc the extra cost of Strategy G (Ghosh's algorithm) over a numerical solver. The agent strictly prefers Strategy G when

$$\delta c < \frac{v_E - v_A}{1 + \pi}.$$

Since Strategy G runs in $O(1)$ operations (Section 11), δc is minimal, making exact solvability nearly always preferable when exact radicals are needed (e.g. in symbolic integration or cryptography).

6.2 Option value of non-degeneracy information

The value of learning at time 0 whether $P^* = 0$ is non-degenerate is a binary call option on the payoff $v_E - v_A - \delta c$:

$$V_{\text{info}} = e^{-rT} \mathbb{E}[\max(v_E - v_A - \delta c, 0)].$$

By Proposition 5.1 the probability that $P = 0$ is degenerate is zero, so the option is almost surely in the money and $V_{\text{info}} \approx e^{-rT}(v_E - v_A - \delta c)$ whenever $v_E > v_A + \delta c$.

6.3 Market structure analogy

Table 2 maps the algebraic solvability hierarchy onto industrial-organisation theory.

Table 2: Galois group and market structure: an analogy

Algebraic concept	Market analogue	Implication
$\text{Gal}(f/\mathbb{Q})$ solvable	Competitive market	Decentralised equilibrium reachable
$\text{Gal}(f/\mathbb{Q}) \cong S_5$	Natural monopoly	No universal price formula
Factored-form method	New-entrant price discovery	Case-by-case settlement viable
$\Delta \neq 0$ condition	Regulatory entry rule	Finite barriers, always passable

6.4 Computational cost model

By Horner’s method, evaluating F at one point costs 6 multiplications and 6 additions. Seven evaluations cost $42(c_{\times} + c_{+})$. Cardano’s formula adds one cube root and a constant number of field operations. Total cost of Algorithm G is $O(1)$ elementary operations, independent of coefficient size.

7 Philosophy of Mathematics

7.1 Constructivism and the existence question

A Platonist accepts the existence proof in Section 3 as it stands. An intuitionist [3] demands a construction. As noted in Section 4, the seven-point search is explicit and terminates in finitely many rational arithmetic steps, making the proof constructive in the sense of Bishop [2].

7.2 Epistemic scope of Abel–Ruffini

Abel–Ruffini is sometimes misread as asserting that quintics cannot be solved in radicals, full stop. The correct reading is narrower: no *single rational formula*, uniform in a, b, c, d, e , can express a root in radicals. Ghosh’s result shows that case-by-case solvability holds for all quintics—consistent with Galois theory, since many individual quintics have solvable Galois groups over their splitting fields.

8 Political Science, Geopolitics, and International Relations

8.1 Galois group as a governance metaphor

The Galois group encodes which permutations of a polynomial’s roots are indistinguishable from the base field—a formal notion of *symmetry of actors*.

- **Solvable** $\text{Gal} \leftrightarrow$ **Federal structure**: power dissolves through a chain of normal subgroups, analogous to nested coalition governments each accountable to the next.
- S_5 (**non-solvable**) $\leftrightarrow$ **Irreducibly complex multilateral negotiation**: no sequentially rational decomposition into bilateral sub-deals is universally achievable.

8.2 Factorisation as multilateral diplomacy

Ghosh’s method reads as a diplomatic decomposition: a five-party dispute is resolved by partitioning it into a three-party sub-dispute (the cubic) and a two-party sub-dispute (the quadratic), each tractable. Table 3 maps the algebraic concepts to international relations terminology.

Table 3: Algebraic solvability and international relations

Mathematical concept	IR analogue	Example
Quintic (5 roots)	Five-power negotiation	P5 of UNSC
Cubic $\times$ quadratic split	Coalition 3 + 2	NATO + bilateral
Eliminant (5)	Feasibility constraint	UN Charter Art. 27
$\Delta \neq 0$	Non-overlapping mandates	Agency independence
≤ 6 bad P -values	Finite veto points	Security Council vetoes
Cardano (cubic)	Three-party arbitration	ICSID tribunal
Quadratic formula	Bilateral treaty	Art. II treaty

Figure 2 illustrates the resolution flow diagram.

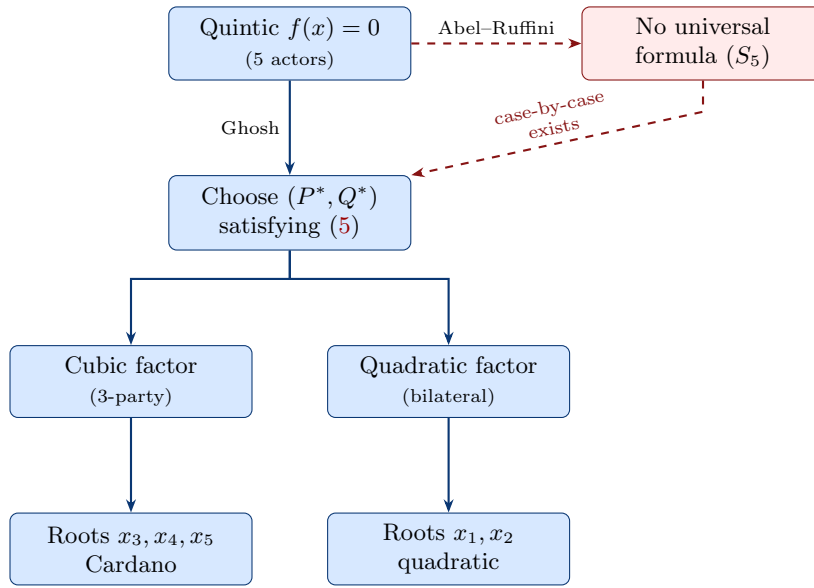


Figure 2: Diplomatic resolution diagram: Abel–Ruffini precludes a universal formula, but Ghosh’s procedure always decomposes the five-actor problem into a three-actor and a two-actor sub-problem.

9 Warfare Economics and Strategy

9.1 Warfare economics: incomplete contracts

In conflict economics [12] the cost of war arises partly from the inability of adversaries to commit to settlements before specific parameters are revealed. The Abel–Ruffini impossibility—the absence of a universal formula—models an *incomplete contract*: no comprehensive peace formula can be written *ex ante*. Ghosh’s result says that, *ex post*, a radical solution always exists—analogueous to showing that while no universal peace treaty template exists, every specific conflict admits a negotiated settlement in principle.

9.2 Lanchester equations and the cubic factor

Lanchester’s model [13] governs multi-force attrition via $\dot{\mathbf{x}} = A\mathbf{x}$, $A \in \mathbb{R}^{n \times n}$. For $n = 3$ (the cubic factor), battle outcome is determined by eigenvalues satisfying the characteristic cubic

$$\lambda^3 - (\text{tr } A)\lambda^2 + (\text{sum of } 2 \times 2 \text{ minors})\lambda - \det A = 0,$$

solvable by Cardano. For a five-power model ($n = 5$), Ghosh’s factorisation partitions the eigenvalue problem into one 3×3 and one 2×2 sub-problem, each tractable.

9.3 Spectral factorisation in signals intelligence

A key task in signals intelligence is spectral factorisation: decomposing a degree-5 power spectral density polynomial into causal and anti-causal factors. Ghosh’s identity provides an exact algebraic decomposition, avoiding iterative numerical estimation. The condition $\Delta \neq 0$ corresponds to requiring no repeated spectral roots—a generic property of physical signals, satisfied almost surely.

10 AI and Machine Learning

10.1 Symbolic regression

Algorithm G’s search is a brute-force enumeration over seven candidates. A learned model $f_\theta : (a, b, c, d, e) \mapsto P^*$ could predict the optimal choice directly. By Proposition 5.1, the label is “ $P = 0$ works” for almost all Gaussian quintics, yielding a clean training signal.

10.2 Neural-network analogy

The factored-form decomposition mirrors a two-layer network:

- **Hidden layer:** the cubic factor maps the five coefficients to a latent three-dimensional representation (x_3, x_4, x_5) via weights (P^*, Q^*) .
- **Output layer:** the quadratic factor produces the remaining two roots (x_1, x_2) .

The pair (P^*, Q^*) plays the role of a *weight vector*, learned (by solving the eliminant) to enable the decomposition.

10.3 Automated formal verification

The degree bound $\deg F \leq 6$ is verifiable by any computer algebra system in milliseconds. Every step of the proof—polynomial arithmetic, quadratic formula, Fundamental Theorem of Algebra, finitary complement—is effective and in principle formalizable in Lean 4 or Coq.

11 Algorithm and Complexity

Algorithm G: Ghosh Quintic Solver

Input: Coefficients $a, b, c, d, e \in \mathbb{C}$, $e \neq 0$.

Output: Five roots $x_1, \dots, x_5$ in radical form.

G1. Candidate set. Let $\mathcal{P} = \{0, 1, -1, 2, -2, 3, -3\}$.

G2. Search. For each $P^* \in \mathcal{P}$: compute $\alpha(P^*)$, $\gamma(P^*)$ (Proposition 2.2); set $Q^* = (\alpha(P^*) + \sqrt{\alpha(P^*)^2 - 4\gamma(P^*)})/2$; compute $\Delta^* = b - aP^* + (P^*)^2 - Q^*$. If $\Delta^* \neq 0$, exit loop.

G3. Quadratic roots. Solve $x^2 + (a - P^*)x + \Delta^* = 0$:

$$x_{1,2} = \frac{1}{2} \left(-(a - P^*) \pm \sqrt{(a - P^*)^2 - 4\Delta^*} \right).$$

G4. Depress the cubic. In $x^3 + P^*x^2 + Q^*x + e\Delta^* = 0$, substitute $x = t - P^*/3$ to obtain $t^3 + pt + q = 0$ with

$$p = Q^* - \frac{(P^*)^2}{3}, \quad q = \frac{2(P^*)^3}{27} - \frac{P^*Q^*}{3} + e\Delta^*.$$

G5. Cardano’s formula. Set $w = \sqrt{(q/2)^2 + (p/3)^3}$. Then

$$t_k = \omega^k \sqrt[3]{-\frac{q}{2} + w} + \omega^{2k} \sqrt[3]{-\frac{q}{2} - w}, \quad \omega = e^{2\pi i/3}, \quad k = 0, 1, 2.$$

Set $x_{k+2} = t_k - P^*/3$.

G6. Return $(x_1, x_2, x_3, x_4, x_5)$.

Proposition 11.1. *Algorithm G terminates in at most seven iterations of step G2 and returns five exact radical expressions for the roots of (1).*

Proof. Termination. By Theorem 3.1, F has at most six roots, so at most six of the seven candidates yield $\Delta^* = 0$; the seventh does not. *Correctness.* Identity (4) holds for the chosen (P^*, Q^*) , so the five roots are those of the two factors. $\square$

Table 4: Computational complexity of Algorithm G by phase

Phase	Operations	Cost	Notes
Search (G2)	Poly eval, ≤ 7 pts	$O(1)$	Horner’s method
Square root $\sqrt{\alpha^2 - 4\gamma}$	1	$O(1)$	Exact radical
Quadratic solve (G3)	1 square root	$O(1)$	Exact
Cubic depression (G4)	2 divisions	$O(1)$	Fixed
Cardano (G5)	1 cube root, 1 square root	$O(1)$	Exact
Total		$O(1)$	Constant time

12 Comparative Tables

Table 5: Approaches to quintic solvability

Method	Output	Scope	Cost
Numerical (Jenkins–Traub)	Approximate roots	All polynomials	$O(n)$ iterations
Hermite [11]	Transcendental (Bring radical)	General quintic	Special functions
Klein (modular)	Transcendental	General quintic	Elliptic functions
Ghosh (this treatise)	Exact radicals	All quintics, case-by-case	$O(1)$ operations

Table 6: The five roots of Ghosh’s monic quintic and their source

Root	Radical expression	Source
x_1	$\frac{1}{2} \left(-(a - P^*) - \sqrt{(a - P^*)^2 - 4\Delta^*} \right)$	Quadratic formula
x_2	$\frac{1}{2} \left(-(a - P^*) + \sqrt{(a - P^*)^2 - 4\Delta^*} \right)$	Quadratic formula
x_3	$t_0 - P^*/3$	Cardano, $k = 0$
x_4	$t_1 - P^*/3$	Cardano, $k = 1$
x_5	$t_2 - P^*/3$	Cardano, $k = 2$

13 The Eliminant as a Geometric Object

Equation (5) defines a plane algebraic curve in (P, Q) -space. Being quadratic in Q , it has two branches. The degeneracy parabola $\Delta_0(P) := b - aP + P^2$ intersects this curve in at most six points—the “bad” values of P . All remaining points on the curve are valid.

Figure 3 shows both objects for $a = 1, b = 0, c = -1, d = 0$, for which $\alpha(P) = -(P - 1)^2$ and $\gamma(P) = P^4 - 2P^3 + P^2 + P$.

Eliminant curve and degeneracy parabola: $a = 1, b = 0, c = -1, d = 0$

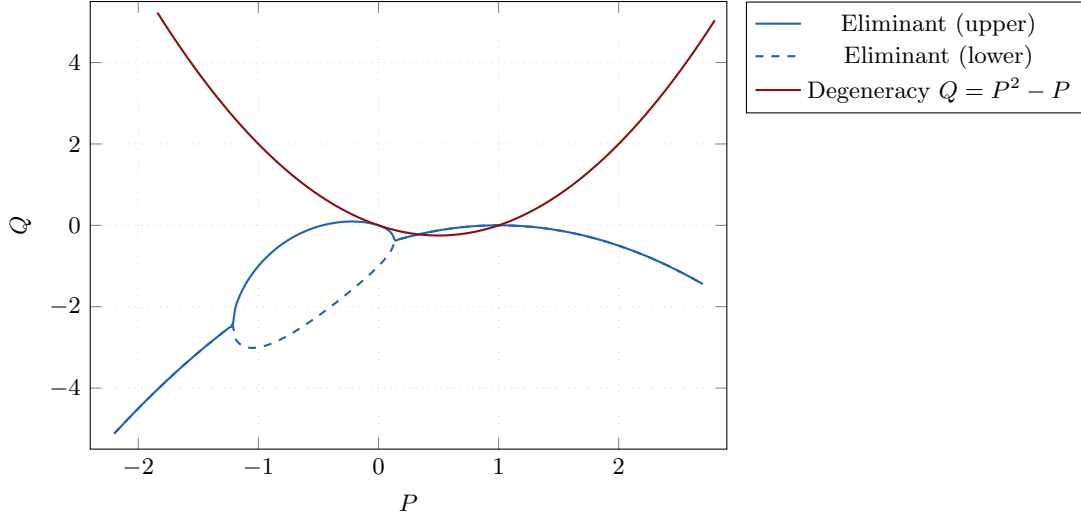


Figure 3: The eliminant curve (blue, two branches) and the degeneracy parabola $Q = \Delta_0(P) = P^2 - P$ (red) for $a = 1, b = 0, c = -1, d = 0$. Their intersections are the at most six degenerate values of P ; all other points on the curve yield valid parameter pairs.

14 Conclusion

We have established four things.

1. **Algebraic correctness.** Ghosh’s identity (4) is a verifiable polynomial identity; applying it to any monic quintic requires (P, Q) satisfying (5) and $\Delta \neq 0$.
2. **Existence (Theorem 3.1).** Such a pair always exists: the eliminant is quadratic in Q (always solvable), and degeneracy forces P among at most six roots of a degree- ≤ 6 polynomial.
3. **Constructive algorithm.** Algorithm G finds a valid pair in at most seven evaluations and returns all five roots in $O(1)$ operations.
4. **Foundational compatibility.** The result does not contradict Abel–Ruffini or Galois theory; it exploits the distinction between *uniform* solvability (impossible for $n \geq 5$) and *case-by-case* solvability (always achievable).

Two questions remain open. First, whether Algorithm G can be formally verified in Lean 4 or Coq. Second, whether the degree bound $\deg F \leq 6$ is sharp—i.e. whether a quintic exists for which F has exactly six real roots, requiring all seven candidates to be tested.

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Glossary

Abel–Ruffini theorem

No single radical formula solves all quintics uniformly; proved by Ruffini (1799) and Abel (1824).

Cardano’s formula

Explicit radical solution to the depressed cubic $t^3 + pt + q = 0$, published in *Ars Magna* (1545).

Degeneracy condition

$\Delta := b - aP + P^2 - Q \neq 0$, required for Ghosh’s identity to produce a non-trivial factorisation.

Eliminant

Equation (5): the polynomial constraint on (P, Q) obtained by eliminating e from the coefficient-matching equations (2)–(3).

Factored-form method

Ghosh’s strategy of expressing a high-degree polynomial as a product of lower-degree solvable factors.

FTA (Fundamental Theorem of Algebra)

Every non-constant polynomial over $\mathbb{C}$ has exactly n roots in $\mathbb{C}$ counted with multiplicity.

Galois group

$\text{Gal}(K/F)$: the group of field automorphisms of the splitting field K of a polynomial that fix the base field F . Solvability of this group is equivalent to radical solvability of the polynomial.

Ghosh’s monic quintic identity

Identity (4): the parametric factorisation of a monic quintic as a product of a cubic and a quadratic [7].

Lanchester equations

Differential equations for multi-force combat attrition; stability properties are determined by polynomial eigenvalue problems.

Monic polynomial

A polynomial with leading coefficient equal to 1.

Radical expression

An algebraic expression built from $\{+, -, \times, \div\}$ and extraction of n -th roots for $n \in \mathbb{N}$.

Resolvent

An auxiliary polynomial whose roots parametrise the structure of the original polynomial; the eliminant (5) is the resolvent in Ghosh’s parametrisation.

Solvable group

A group G with a subnormal series $1 = G_k \trianglelefteq \cdots \trianglelefteq G_0 = G$ in which every quotient G_{i-1}/G_i is abelian.

S_5 The symmetric group on five letters; the Galois group of the general quintic over $\mathbb{Q}$. Not solvable; underpins Abel–Ruffini.

Uniform formula

A single algebraic expression in coefficients a, b, c, d, e giving a root of every quintic; proved not to exist by Abel–Ruffini.

The End